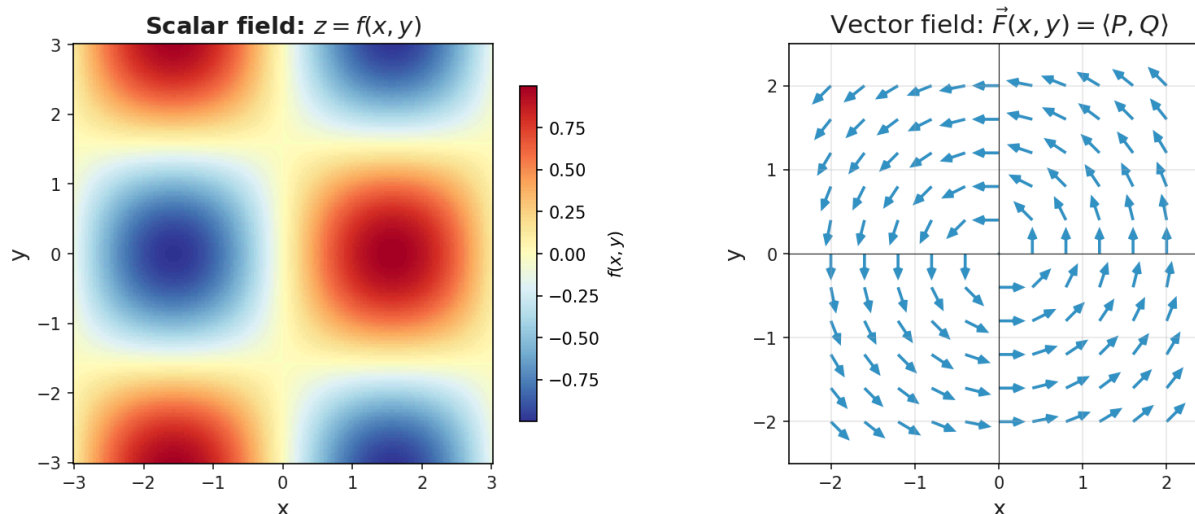


From Scalar Fields to Vector Fields



A **scalar field** assigns a **number** to each point. A **vector field** assigns a **vector** to each point.

Definition of a Vector Field

Vector Field (2D)

$$\vec{F}(x, y) = \langle P(x, y), Q(x, y) \rangle$$

Vector Field (3D)

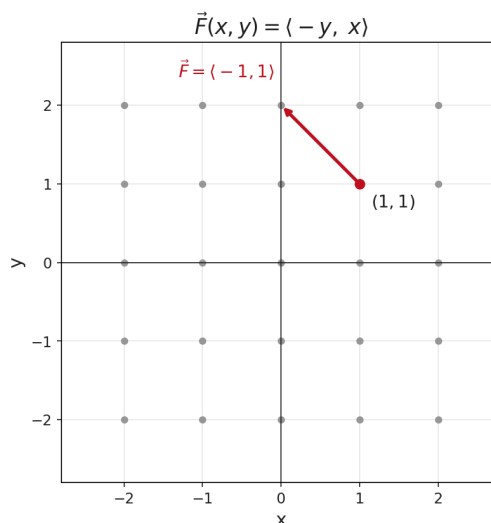
$$\vec{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle$$

The functions P , Q , R are the **component functions** — ordinary scalar functions that give each component of the vector at each point.

Example 1: Plot a Vector Field

Example 1

Plot the vector field $\vec{F}(x, y) = \langle -y, x \rangle$ by evaluating $\vec{F}$ at several points and drawing the vectors.



At $(1, 1)$: $\vec{F} = \langle -1, 1 \rangle$. Draw an arrow from $(1, 1)$ pointing left and up.

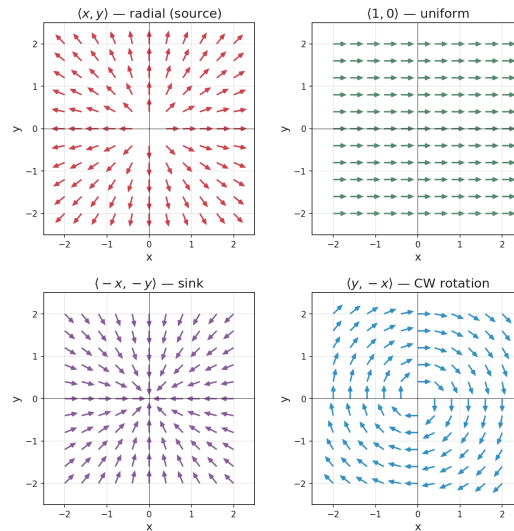
Now you try:

- At $(1, 0)$: $\vec{F} = \langle ? , ? \rangle$
- At $(0, 1)$: $\vec{F} = \langle ? , ? \rangle$
- At $(-1, 0)$: $\vec{F} = \langle ? , ? \rangle$
- At $(2, 0)$: $\vec{F} = \langle ? , ? \rangle$

$\langle 0, 1 \rangle$, $\langle -1, 0 \rangle$, $\langle 0, -1 \rangle$, $\langle 0, 2 \rangle$. Do you see a pattern forming?

Exploring 2D Vector Fields

2D Vector Field Plotter (GeoGebra) [<https://www.geogebra.org/material/iframe/id/cXgNb58T>] — Interactive tool for plotting 2D vector fields. Enter component functions $P(x,y)$ and $Q(x,y)$ to see the field.



Radial field: $\langle x, y \rangle$ — every arrow points away from the origin. A **source**.

Uniform field: $\langle 1, 0 \rangle$ — identical arrows everywhere. A constant wind.

Sink field: $\langle -x, -y \rangle$ — every arrow points toward the origin.

Clockwise rotation: $\langle y, -x \rangle$ — the mirror image of Example 1.

Vector Fields in 3D

3D Vector Field Plotter (GeoGebra) [<https://www.geogebra.org/m/u3xregNW>] — Interactive tool for plotting 3D vector fields. Enter component functions $P(x,y,z)$, $Q(x,y,z)$, and $R(x,y,z)$ to see the field in 3D. Rotate to explore.

Try it: $\langle x, y, -z \rangle$ — a “fountain” field. Water flows outward from the center and falls back down.

Physics examples:

- **Gravity:** $\vec{F} = -\frac{mMG}{|\vec{r}|^3} \vec{r}$ (points toward mass, inverse-square)
- **Electric field:** $\vec{E} = \frac{q}{|\vec{r}|^3} \vec{r}$ (same structure, points away from $+$ charge)

Gradient Fields and Conservative Vector Fields

Gradient Field / Conservative Vector Field

A vector field $\vec{F}$ is called a **gradient field** (or **conservative**) if there exists a scalar function f such that:

$$\vec{F} = \nabla f = \langle f_x, f_y \rangle \quad (\text{or } \langle f_x, f_y, f_z \rangle \text{ in 3D})$$

The function f is called a **potential function** for $\vec{F}$.

Why “conservative”? A conservative force does zero net work around any closed path. Gravity is conservative — energy spent going uphill is recovered coming back down.

Not every vector field is conservative. A big question in Chapter 16: *which fields are conservative, and how can we tell?*

Example 2: Verify a Gradient Field

Example 2

Show that $\vec{F}(x, y) = \langle 2xy + 3, x^2 - 4y \rangle$ is a gradient field by finding a potential function f .

We need f with $\nabla f = \vec{F}$, i.e. $f_x = 2xy + 3$ and $f_y = x^2 - 4y$.

How Gradient Fields Look Different

Gradient field ($\vec{F} = \nabla f$): arrows are **perpendicular to level curves** of f , pointing from low to high values. No swirling — a leaf dropped in the flow moves but never spins.

Non-gradient field: arrows can form closed loops, swirl, or rotate. The rotation field $\langle -y, x \rangle$ from Example 1 is the classic case.

Visual test: If a vector field has arrows forming closed loops, it is **not conservative**. Conservative fields “flow downhill” from a potential function.

In Section 16.3, we'll develop precise tests for when a field is conservative and systematic methods for finding the potential function. For now, the gradient concept is what matters.

Homework

Section 16.1: 3, 7, 13--22, 25, 27, 29, 31--34